

<i>Series A</i>	<i>Series B</i>	<i>Series C</i>
200	200	1
200	205	989
200	202	2
200	203	3
200	190	5
Mean = 200	200	200

Measure of Dispersion

❖ *Scatteredness (homogeneity or heterogeneity)*

The degree to which numerical data tend to spread about an average value is called variation or dispersion of data.

Measures of Dispersion: Range
 Quartile Deviation
 Mean Deviation
 Standard Deviation

Absolute Measure of Dispersion:

Relative Measure of Dispersion:

Range

Definition: Difference between the value of the smallest item and the value of the largest item in the distribution.

$$\text{Range} = L - S$$

L – Largest Value, S- Smallest Value

The *relative measure* corresponding to range is called the ***coefficient of range***,

$$\text{Coefficient of Range} = \frac{L-S}{L+S}$$

Example 1: The following are the prices of shares of a company from Monday to Saturday:

Days	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Price(Rs.)	200	210	208	160	220	250

Calculate the range and its coefficient.

Solution: $\text{Range} = L - S = 250 - 160 = 90$

$\text{Range} = \text{Rs. } 90$

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{250-160}{250+160} = \frac{90}{410} = 0.22$$

In a frequency distribution, range is calculated by taking the difference between the lower limit of the lowest class and the upper limit of the highest class.

Example 2:

Marks	10-20	20-30	30-40	40-50	50-60	60-70
No. of Students	12	18	27	20	17	6

$$\text{Range} = L - S = 70 - 10 = 60$$

$$\text{Coefficient of Range} = \frac{L-S}{L+S} = \frac{70-10}{70+10} = \frac{60}{80} = 0.75$$

Quartile Deviation

Definition: Average amount by which the two quartiles differ from the median.

$$\text{Quartile Deviation (Q.D.)} = \frac{Q_3 - Q_1}{2}$$

- The Median $\pm$ Q.D. covers exactly 50 per cent of the observations.
- When Q.D. is very small, it describes high uniformity or small variation of the central 50% items, and a high Q.D. means that the variation among the central items is large.

Relative measure of Q.D.

$$\text{Coefficient of Q.D.} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

It can be used to compare the degree of variation in different distributions.

Example 3: Calculate the value of Q.D. and its coefficient of Q.D. from the following data.

Roll No.	1	2	3	4	5	6	7
Marks	20	28	40	12	30	15	50

Solution: Marks in ascending order 12 15 20 28 30 40 50

$$Q_1 = \text{Size of } \frac{N+1}{4} \text{ th item} = \text{Size of } \frac{7+1}{4} = 2^{\text{nd}} \text{ item.}$$

Size of 2nd item is 15. Hence $Q_1 = 15$

$$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{ th item} = \text{Size of } 3\left(\frac{7+1}{4}\right) = 6^{\text{th}} \text{ item.}$$

Size of 6th item is 40. Hence $Q_3 = 40$.

$$\therefore Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 15}{2} = 12.5$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 15}{40 + 15} = 0.455$$

Example 4: Compute the value of Q.D. and its coefficient from the following data.

Marks	10	20	30	40	50	60
No. of Students	4	7	15	8	7	2

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>cumulative frequency</i>
10	4	4
20	7	11
30	15	26
40	8	34
50	7	41
60	2	43

$$Q_1 = \text{Size of } \frac{N+1}{4} \text{ th item} = \text{Size of } \frac{43+1}{4} = 11^{\text{th}} \text{ item.}$$

Size of 11^{th} item is 20. Hence **$Q_1 = 20$**

$$Q_3 = \text{Size of } 3\left(\frac{N+1}{4}\right) \text{ th item} = \text{Size of } 3\left(\frac{43+1}{4}\right) = 33^{\text{rd}} \text{ item.}$$

Size of 33^{rd} item is 40. Hence **$Q_3 = 40$** .

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{40 - 20}{2} = 10$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{40 - 20}{40 + 20} = 0.333$$

Example 4: Compute the value of Q.D. and coefficient of Q.D. from the following data

<i>C.I.</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80
<i>f</i>	12	19	5	10	9	6	6

Solution:

<i>Marks</i>	<i>Frequency</i>	<i>Cumulative Frequency</i>
10-20	12	12
20-30	19	31
30-40	5	36
40-50	10	46
50-60	9	55
60-70	6	61
70-80	6	67
	<i>$N = 67$</i>	

$$Q_1 = \text{Size of } \frac{N}{4} \text{ th item} = \text{Size of } \frac{67}{4} = 16.75^{\text{th}} \text{ item.}$$

Q_1 lies in the interval **20-30**

$$Q_1 = l + \frac{\frac{N}{4} - \text{c.f.}}{f} \times i \quad l = 20, N/4 = 16.75, \text{c.f.} = 12 \quad f = 19, i = 10$$

$$Q_1 = 20 + \frac{\frac{67}{4} - 12}{19} \times 10 = 20 + 2.5 = 22.5$$

Hence $Q_1 = 22.5$

$$Q_3 = \text{Size of } \frac{3N}{4} \text{ th item} = \text{Size of } \frac{3 \times 67}{4} = 50.25^{\text{th}} \text{ item.}$$

Q_3 lies in the class **50-60**.

$$Q_3 = l + \frac{\frac{3N}{4} - \text{c.f.}}{f} \times i \quad l = 50, 3N/4 = 50.25, \text{c.f.} = 46 \quad f = 9, i = 10$$

$$Q_3 = 50 + \frac{50.25 - 46}{9} \times 10 = 50 + 4.72 = 54.72$$

$$\text{Hence } Q_3 = 54.72$$

$$Q.D. = \frac{Q_3 - Q_1}{2} = \frac{54.72 - 22.5}{2} = 16.11$$

$$\text{Coefficient of } Q.D. = \frac{Q_3 - Q_1}{Q_3 + Q_1} = \frac{54.72 - 22.5}{54.72 + 22.5} = 0.4172$$

Mean Deviation

Definition: M.D. is the average difference between the observations in a distribution and the *median* or *mean* of that series.

$$\text{Mean Deviation or M.D.} = \frac{\sum |D|}{N}$$

Where $|D|$ is the deviations from ***median*** ignoring signs.

For individual observations:

- (i) Compute median of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D|$.
- (iii) Obtain the total of these observations, $\sum |D|$.
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of ***mean deviation***.

Relative Measure of M.D. :

$$\text{Co-efficient of M.D.} = \frac{M.D.}{Median}$$

Example 5: Calculate the mean deviation of the two income groups.

I (Rs.)	II (Rs.)
4000	3000
4200	4000
4400	4200
4600	4400
4800	4600
	4800
	5800

Solution:

Group I		Group II	
Rs.	$ D $	Rs.	$ D $
4000	400	3000	1400
4200	200	4000	400
4400	0	4200	200
4600	200	4400	0
4800	400	4600	200
		4800	400
		5800	1400
N = 5	$\Sigma D = 1200$	N = 7	$\Sigma D = 4000$

Mean deviation : **I group** M.D. = $\frac{\Sigma|D|}{\sqrt{N}}$

Median = $\frac{\sqrt{N}+1}{2}$ *th* item = $\frac{5+1}{2} = 3^{\text{rd}}$ item. Size of the 3^{rd} item = 4400.

$$\text{M.D.} = \frac{1200}{5} = \mathbf{240}$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 240**.

Mean deviation : **II group** M.D. = $\frac{\sum |D|}{N}$

$$\text{Median} = \frac{N+1}{2} \text{th item} = \frac{7+1}{2} = 4^{\text{th}} \text{ item.}$$

Size of the 4th item = 4400.

$$\text{M.D.} = \frac{4000}{5} = 571.43$$

i.e., the average deviation of the individual incomes from the median income is **Rs. 571.43**.

$$\text{Co-efficient of M.D. (I - Group)} = \frac{M.D.}{Median} = \frac{240}{4400} = 0.055$$

$$\text{(II- Group)} = \frac{571.43}{4400} = 0.13$$

Mean deviation – Discrete Series

$$\text{M.D.} = \frac{\sum f|D|}{N}$$

- (i) Compute median of the series.
- (ii) Take deviations of items from median ignoring $\pm$ signs and denote these deviations by $|D|$.
- (iii) Multiply these deviations by the respective frequencies and Obtain the total, $\sum f|D|$.
- (iv) Divide the total obtained in step (iii) by the number of observations to get the value of ***mean deviation***.

Example 6: The number of telephone calls received at an exchange in 245 successive one-minute intervals are shown in the following frequency distribution. Compute the mean deviation about the median.

<i>Number of Calls</i>	0	1	2	3	4	5	6	7
<i>Frequency</i>	14	21	25	43	51	40	39	12

Solution:

<i>No. of Calls</i>	<i>f</i>	<i>c.f.</i>	<i> D </i>	<i>f D </i>
0	14	14	4	56
1	21	35	3	63
2	25	60	2	50
3	43	103	1	43
4	51	154	0	0
5	40	194	1	40
6	39	233	2	78
7	12	245	3	36
	<i>N = 245</i>			$\Sigma f D = 366$

Median = Size of $\frac{N+1}{2}$ th item = $\frac{245+1}{2} = 123^{\text{rd}}$ item .

Hence the median value is 4.

$$\text{M.D.} = \frac{\Sigma f|D|}{N} = \frac{366}{245} = \mathbf{1.49}$$

In Continuous Series:

We have to obtain the mid-points of the various classes and take the deviations of these **mid-points** from median.

Calculate the coefficient of mean deviation from the following data:

<i>Marks</i>	10-20	20-30	30-40	40-50	50-60	60-70	70-80	80-90
<i>No. of Students</i>	2	6	12	18	25	20	10	7

Mean deviation = 12.94

Standard deviation

For the frequency distribution $x_i \mid f_i$; $i = 1, 2, \dots, n$,

$$\text{Variance} = \sigma^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{1}{N} \sum x_i^2 - \left(\frac{1}{N} \sum x_i \right)^2 = \frac{1}{N} \sum x_i^2 - \bar{x}^2$$

(ii) Discrete or Continuous frequency distribution

$$\begin{aligned} \sigma^2 &= \frac{1}{N} \sum f_i (x_i - \bar{x})^2 \\ &= \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2 \text{ or } \frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \end{aligned}$$

$$\text{Standard deviation} = \sqrt{\text{variance}} \quad \sigma = \sqrt{\frac{1}{N} \sum (x_i - \bar{x})^2} \quad \text{or} \quad \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

$\bar{x}$ – Arithmetic mean of the distribution

Coefficient of Variation

Coefficient of Variation : C.V. = $\frac{\sigma}{\bar{x}}$ x 100 (Relative Measure)

- 3) The score of two players A and B in ten innings during a certain season are:

A	32	28	47	63	71	39	10	60	96	14
B	19	31	48	53	67	90	10	62	40	80

Find which of the two players A, B is more consistent in scoring.

Solution:

Calculation of Coefficient of Variation

X	$(X - \bar{X})$	$(X - \bar{X})^2$
32	-14	196
28	-18	324
47	+1	1
63	+17	289
71	+25	625
39	-7	49
10	-36	1296
60	+14	196
96	+50	2500
14	-32	1024
$\sum X = 460$	0	6500

$$\bar{X} = \frac{460}{10} = 46$$
$$\sigma_A^2 = \frac{\sum (x_i - \bar{x})^2}{N} = \frac{6500}{10} = 650$$

Y	$(Y - \bar{Y})$	$(Y - \bar{Y})^2$
19	-31	961
31	-19	361
48	-2	4
53	+3	9
67	+17	289
90	+40	1600
10	-40	1600
62	+12	144
40	-10	100
80	+30	900
$\sum Y = 500$	0	5968

$$\bar{Y} = \frac{500}{10} = 50$$
$$\sigma_B^2 = \frac{\sum (y_i - \bar{y})^2}{N} = \frac{5968}{10} = 596.8$$

$$\sigma_A = \sqrt{\frac{\sum(x_i - \bar{x})^2}{N}} = 25.5$$

$$\sigma_B = \sqrt{\frac{\sum(y_i - \bar{y})^2}{N}} = 24.43$$

$$\text{C.V.}_{(A)} = \frac{\sigma_A}{\bar{x}} \times 100 = 55.43$$

$$\text{C.V.}_{(B)} = \frac{\sigma_B}{\bar{y}} \times 100 = 48.86$$

$$\sum X = 460 ; \quad \sum (X_i - \bar{x}) = 0 ; \quad \sum (X_i - \bar{x})^2 = 6500$$

$$\sum Y = 500 ; \quad \sum (Y_i - \bar{y}) = 0 ; \quad \sum (Y_j - \bar{y})^2 = 5968$$

$$\sigma_A = 25.5$$

$$\sigma_B = 24.43$$

$$\text{C.V.}_{(A)} = 55.43$$

$$\text{C.V.}_{(B)} = 48.86$$

- 4) Suppose that samples of polythene bags from two manufacturers, A and B, are tested by a prospective buyer for bursting pressure, with the following results:

<i>Bursting Pressure (lb.)</i>	<i>Number of Bags</i>	
	<i>A</i>	<i>B</i>
5.0 – 9.9	2	9
10.0 – 14.9	9	11
15.0 – 19.9	29	18
20.0 – 24.9	54	32
25.0 – 29.9	11	27
30.0 – 34.9	5	13

Which set of bags has the highest average burning pressure?

Which has more uniform pressure? If prices are the same,

which manufacturer's bags would be preferred by the buyer? **Why?**

For Manufacturer A

Bursting Pressure (lb.)	<i>m</i>	<i>f</i>	$\left(\frac{m - 17.45}{5}\right)$ <i>d</i>	<i>fd</i>	<i>fd</i>²
4.95-9.95	7.45	2	-2	-4	4
9.95-14.95	12.45	9	-1	-9	9
14.95-19.95	17.45	29	0	0	0
19.95-24.95	22.45	54	1	54	54
24.95-29.95	27.45	11	2	22	44
29.95-34.95	32.45	5	3	15	45
		N = 110		$\Sigma fd=78$	$\Sigma fd^2=160$

$$\bar{X}_A = A + \frac{\sum_{i=1}^n f_i d_i}{N} \times i$$

Here, $A = 17.45$, $\sum f d = 78$, $N = 110$, $i = 5$

$$\bar{X}_A = 17.45 + \frac{78}{110} \times 5 = 21$$

$$\begin{aligned} \sigma_A &= \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times i \\ &= \sqrt{1.455 - 0.503} \times 5 = 4.88 \end{aligned}$$

$$\text{C.V.} = \frac{\sigma_A}{\bar{x}} \times 100 = 23.24\%$$

For Manufacturer B

Bursting Pressure (lb.)	<i>m</i>	<i>f</i>	$\left(\frac{m - 17.45}{5}\right)$ <i>d</i>	<i>fd</i>	<i>fd</i>²
4.95-9.95	7.45	9	-2	-18	36
9.95-14.95	12.45	11	-1	-11	11
14.95-19.95	17.45	18	0	0	0
19.95-24.95	22.45	32	+1	+32	32
24.95-29.95	27.45	27	+2	+54	108
29.95-34.95	32.45	13	+3	+39	117
		N = 110		$\Sigma fd = 96$	$\Sigma fd^2 = 304$

$$\bar{X}_B = A + \frac{\sum_{i=1}^n f_i d_i}{N} \times i$$

Here, $A = 17.45$, $\sum f d = 96$, $N = 110$, $i = 5$

$$\bar{X}_B = 17.45 + \frac{96}{110} \times 5 = 21.81$$

$$\begin{aligned} \sigma_B &= \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \times i \\ &= \sqrt{2.764 - 0.762} \times 5 = 7.075 \end{aligned}$$

$$\text{C.V.} = \frac{\sigma_B}{\bar{x}} \times 100 = 32.44\%$$

$$\bar{X}_A = 21$$

$$\bar{X}_B = 21.81$$

$$\sigma_A = 4.88$$

$$\sigma_B = 7.07$$

$$\text{C.V.}_{(A)} = 23.24\%$$

$$\text{C.V.}_{(B)} = 32.44\%$$

Since the average bursting pressure is higher for manufacturer B, the bags of manufacturer B have higher bursting pressure.